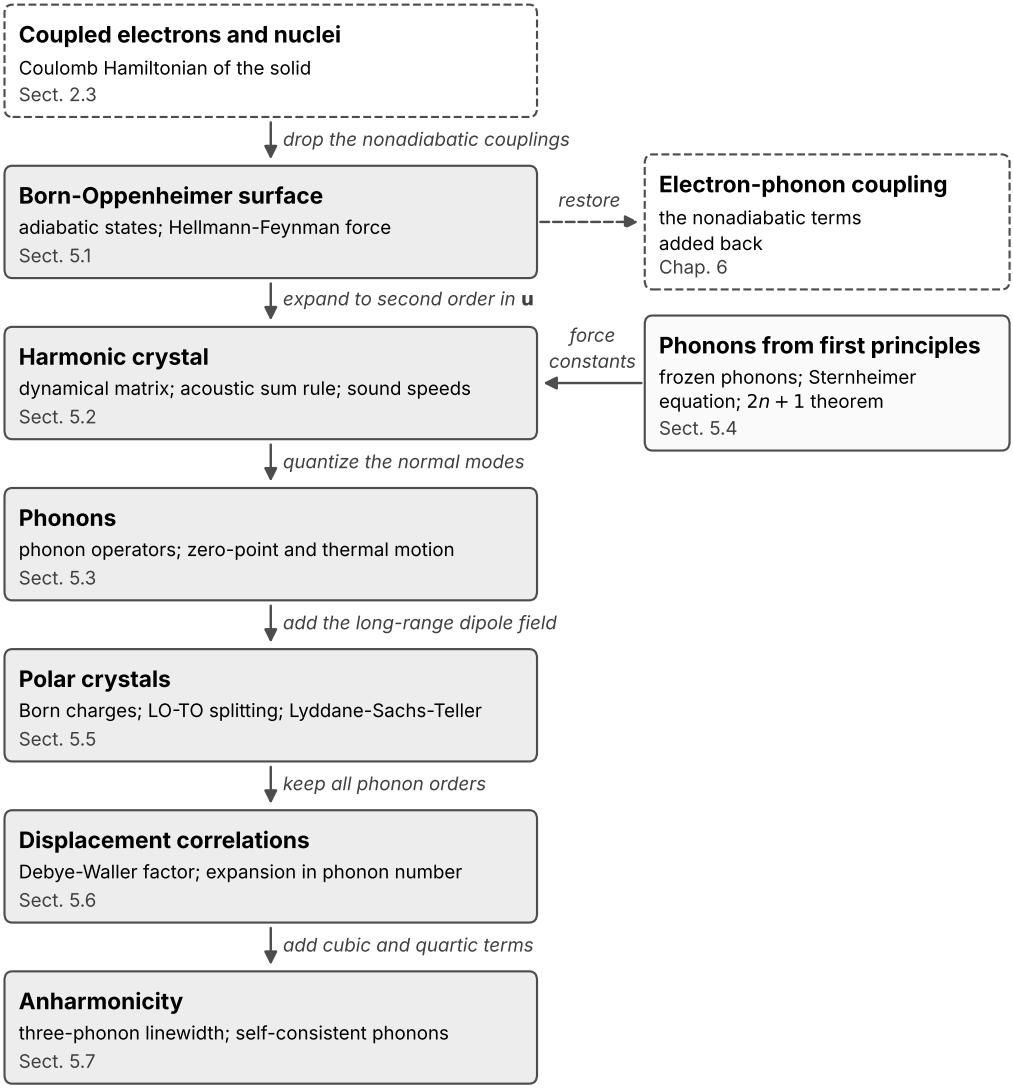
Coupled electrons and nuclei
Coulomb Hamiltonian of the solid
Sect. 2.3
drop the nonadiabatic couplings
Born-Oppenheimer surface
adiabatic states; Hellmann-Feynman force
Sect. 5.1
restore
Electron-phonon coupling
the nonadiabatic terms added back
Chap. 6
expand to second order in u
Harmonic crystal
dynamical matrix; acoustic sum rule; sound speeds
Sect. 5.2
force constants
Phonons from first principles
frozen phonons; Sternheimer equation; 2n + 1 theorem
Sect. 5.4
quantize the normal modes
Phonons
phonon operators; zero-point and thermal motion
Sect. 5.3
add the long-range dipole field
Polar crystals
Born charges; LO-TO splitting; Lyddane-Sachs-Teller
Sect. 5.5
keep all phonon orders
Displacement correlations
Debye-Waller factor; expansion in phonon number
Sect. 5.6
add cubic and quartic terms
Anharmonicity
three-phonon linewidth; self-consistent phonons
Sect. 5.7